

Weakly Contractive Mappings and Fixed Point Theorems

Pathak: An Introduction to Nonlinear Analysis and Fixed Point Theory (Ch. 5, §5.1)

Lecture Notes

2024

Outline

- 1 Introduction: Classical Fixed Point Theorems
- 2 Kannan and Chatterjea Contractions
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Fixed Point Theorem: Classical Formulation

Definition (Fixed Point)

Let $T : X \rightarrow X$ be a mapping. An element $u \in X$ is called a fixed point of T if

$$T(u) = u.$$

Theorem (Banach Contraction Mapping Principle)

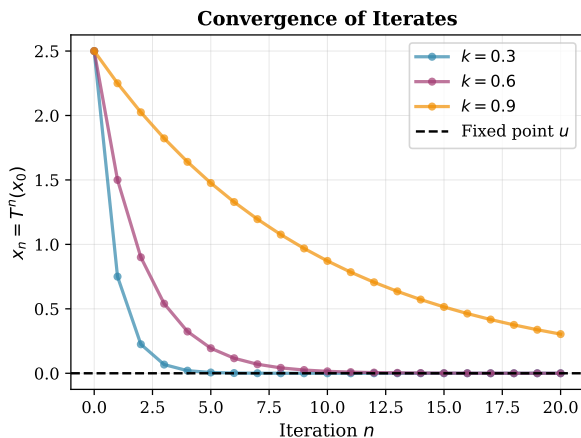
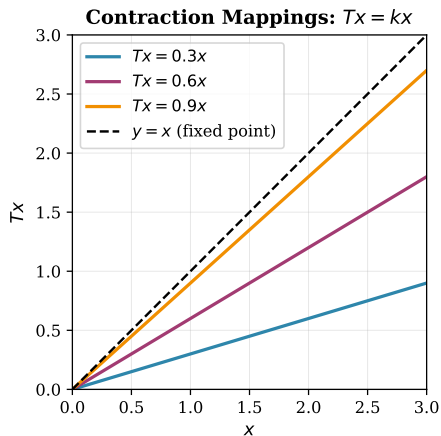
Let (X, d) be a complete metric space and $T : X \rightarrow X$ a mapping such that there exists $\lambda \in [0, 1)$ with

$$d(Tx, Ty) \leq \lambda d(x, y) \quad \forall x, y \in X.$$

Then:

- 1 T has a unique fixed point $u \in X$;
- 2 For any $x_0 \in X$, the sequence $\{T^n(x_0)\}$ converges to u ;
- 3 $d(T^n(x_0), u) \leq \frac{\lambda^n}{1-\lambda} d(x_0, T(x_0)).$

Illustration: Contraction Mapping Principle



Left: Different contraction constants $k = 0.3, 0.6, 0.9$ show how smaller constants yield steeper contractions.

Right: Convergence rates of iterates $x_n = T^n(x_0)$ depend on the contraction constant.

Kannan's Contraction

Theorem (Kannan, 1968)

Let (X, d) be a complete metric space and $T : X \rightarrow X$ such that there exists $r \in [0, \frac{1}{2})$ with

$$d(Tx, Ty) \leq r[d(Tx, y) + d(Ty, x)] \quad (KC)$$

for all $x, y \in X$. Then T has a unique fixed point $u \in X$.

Key Observation: Kannan's condition involves *mixed distances*:

- $d(Tx, y)$: distance from image of x to y
- $d(Ty, x)$: distance from image of y to x

Advantage: T need not be continuous at the fixed point.

Example: For $X = \mathbb{R}$ with usual metric, define

$$T(x) = \begin{cases} 0 & \text{if } x \in (-\infty, 2], \\ \frac{1}{2} & \text{if } x \in (2, +\infty). \end{cases}$$

Then T satisfies Kannan's contraction with $r = \frac{1}{2}$ but is discontinuous at $x = 2$.

Chatterjea's Contraction

Theorem (Chatterjea, 1972)

Let (X, d) be a complete metric space and $T : X \rightarrow X$ such that there exists $r \in [0, \frac{1}{2})$ with

$$d(Tx, Ty) \leq r[d(Tx, y) + d(Ty, x)] \quad (CHC)$$

for all $x, y \in X$. Then T has a unique fixed point $u \in X$.

Remark: Chatterjea's condition has the same form as Kannan's! The *difference* lies in the interpretation and context of application.

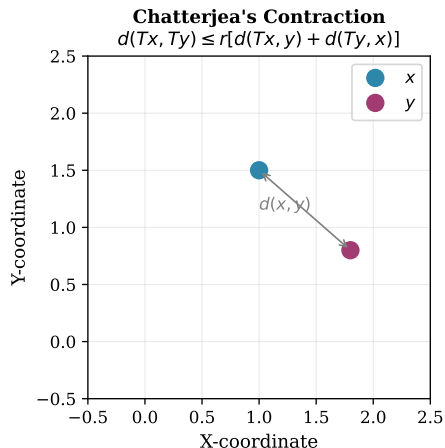
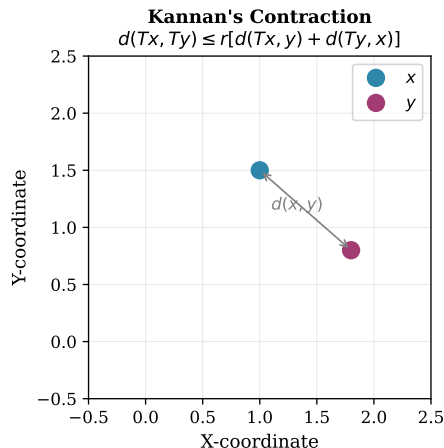
Theorem (Rhoades, 1977)

The contractive conditions:

- Banach's contraction (BC)
- Kannan's contraction (KC)
- Chatterjea's contraction (CHC)

are **independent** i.e. none implies the others in general

Geometric Comparison: Kannan and Chatterjea



Both contractions use mixed distance terms. Kannan's and Chatterjea's conditions relax the requirement of linearity found in Banach contractions, allowing non-continuous mappings to have fixed points.

Meir-Keeler Contraction

Theorem (Meir-Keeler, 1969)

Let (X, d) be a complete metric space and $T : X \rightarrow X$ such that for every $\varepsilon > 0$ there exists $\delta > 0$ with

$$\varepsilon \leq d(x, y) < \varepsilon + \delta \implies d(Tx, Ty) < \varepsilon \quad (MK)$$

Then T is contractive, i.e., $d(Tx, Ty) < d(x, y)$ for all $x \neq y$.

Consequence: If T is contractive and (X, d) is complete, then T has a unique fixed point.

Key Property: Meir-Keeler contractions are *contractive* (strict inequality) but need not be Lipschitz contractions (no linear constant λ).

Interpretation: For any positive distance threshold ε , there exists a “band” of widths δ such that distances in this band are reduced below ε .

Theorem (Reich, 1971)

Let (X, d) be a complete metric space and $T : X \rightarrow X$ such that there exist $a, b, c \in [0, 1)$ with $a + b + c < 1$ and

$$d(Tx, Ty) \leq a d(x, y) + b d(x, Tx) + c d(y, Ty) \quad (RC)$$

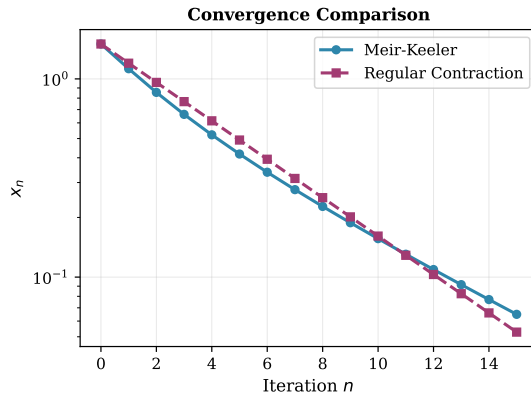
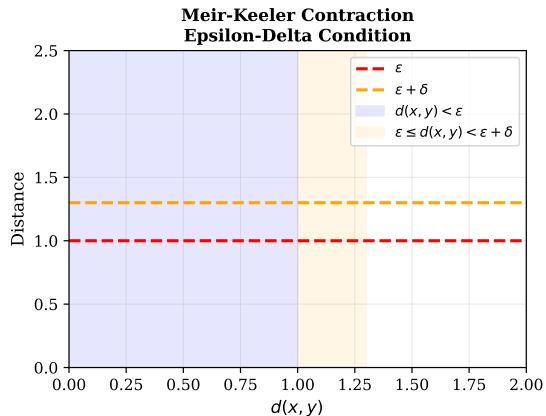
for all $x, y \in X$. Then T has a unique fixed point $u \in X$.

Generalization: Reich's condition combines:

- Distance between x and y (Banach-type term)
- Orbit information: distances $d(x, Tx)$ and $d(y, Ty)$

Parameter Space: The feasible region is $\{(a, b, c) : a, b, c \geq 0, a + b + c < 1\}$.

Comparison: Meir-Keeler and Reich



Left: Meir-Keeler's epsilon-delta condition creates flexible contraction behavior. **Right:** Convergence comparison shows how different contraction types affect the rate of convergence (note log scale).

Čirić Contraction

Theorem (Čirić, 1971)

Let (X, d) be a complete metric space and $T : X \rightarrow X$ such that there exist $a, b, c, e \in [0, 1)$ with $a + b + c + 2e < 1$ and

$$d(Tx, Ty) \leq a d(x, y) + b d(x, Tx) + c d(y, Ty) + e[d(x, Ty) + d(y, Tx)]$$

for all $x, y \in X$. Then T has a unique fixed point $u \in X$.

Generalization: Čirić adds two new terms:

- $d(x, Ty)$: mixed distance from x to image of y
- $d(y, Tx)$: mixed distance from y to image of x

Strictest Condition: The constraint $a + b + c + 2e < 1$ is the most restrictive among the contractions discussed so far.

Hardy-Rogers Contraction

Theorem (Hardy-Rogers, 1973)

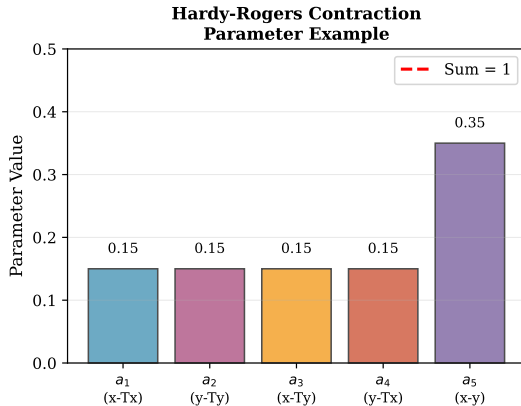
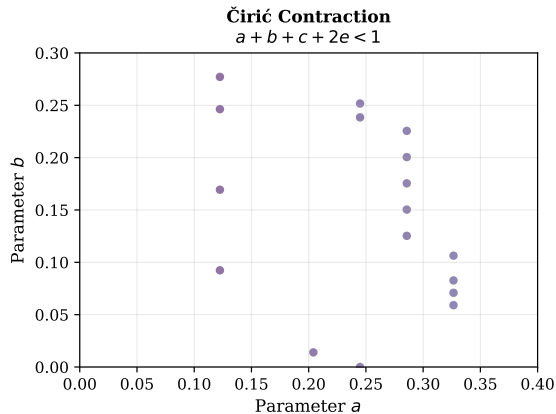
Let (X, d) be a complete metric space and $T : X \rightarrow X$ such that there exist $a_1, a_2, a_3, a_4, a_5 \in [0, 1)$ with $\sum_{j=1}^5 a_j < 1$ and

$$\begin{aligned} d(Tx, Ty) \leq & a_1 d(x, Tx) + a_2 d(y, Ty) + a_3 d(x, Ty) \\ & + a_4 d(y, Tx) + a_5 d(x, y) \end{aligned}$$

for all $x, y \in X$. Then T has a unique fixed point $u \in X$.

Most General: Five independent parameters allow maximum flexibility in defining the contraction behavior.

Čirić and Hardy-Rogers Comparison



Left: Parameter space for Čirić contractions showing feasible regions. **Right:** Example of Hardy-Rogers parameter values. The constraint $\sum a_i < 1$ ensures convergence while allowing flexible weighting of different distance measures.

Zamfirescu Contraction

Theorem (Zamfirescu, 1972)

Let (X, d) be a complete metric space and $T : X \rightarrow X$ such that

$$d(Tx, Ty) \leq \max \left\{ d(x, y), \frac{1}{2}[d(x, Tx) + d(y, Ty)], \frac{1}{2}[d(x, Ty) + d(y, Tx)] \right\}$$

for all $x, y \in X$. Then T has a unique fixed point $u \in X$.

Key Feature: Uses *maximum* of three distance expressions, providing a non-linear contractive condition.

Flexibility: The mapping contracts relative to whichever distance measure is largest, allowing asymmetric behavior.

Berinde Contraction

Theorem (Berinde, 2004)

Let (X, d) be a complete metric space and $T : X \rightarrow X$ such that there exist $r \in [0, 1)$ and $L \geq 0$ with

$$d(Tx, Ty) \leq r d(x, y) + L d(y, Tx)$$

for all $x, y \in X$. Then T has a unique fixed point $u \in X$.

Quasi-contraction: Berinde's condition involves both:

- A standard contraction term: $r d(x, y)$
- An auxiliary term: $L d(y, Tx)$

Application: Useful in cases where the linear term alone does not guarantee contraction.

Bessaga's Converse Theorem

Theorem (Bessaga, 1959)

Let X be an arbitrary set and $T : X \rightarrow X$ such that T^n has a unique fixed point $u \in X$ for every $n \geq 1$. Then for every $\lambda \in (0, 1)$, there exists a metric $d = d_\lambda$ on X such that:

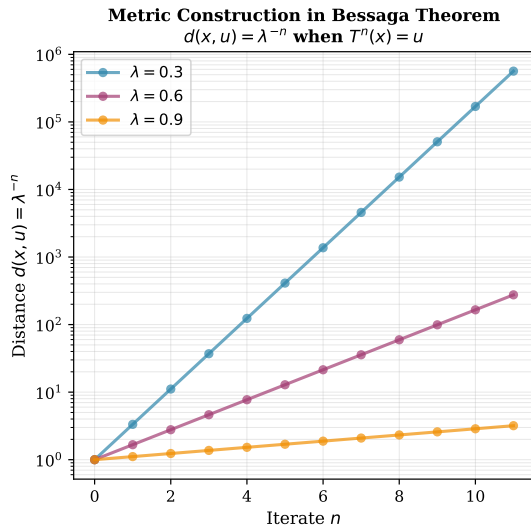
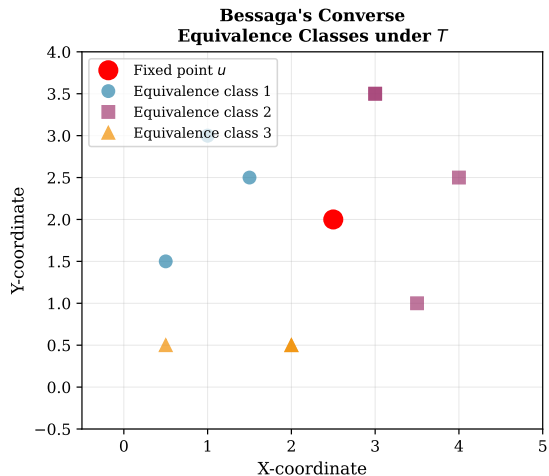
- 1 (X, d) is a complete metric space;
- 2 T is a contraction on (X, d) with Lipschitz constant λ .

Significance: Shows that the Banach contraction principle is *complete* — having unique fixed points for all iterates is equivalent to the existence of a metric making T a contraction.

Construction Key:

- Partition X into equivalence classes under the relation T
- Define distance $d(x, u) = \lambda^{-n}$ if $T^n(x) = u$

Illustration: Bessaga's Converse



Left: Equivalence classes under the action of T partition the space. Each class has points related by iteration.

Definition (ε – Chain)

Let (X, d) be a metric space and $\varepsilon > 0$. A finite sequence $x_0, x_1, \dots, x_n$ of points in X is called an ε -chain joining x_0 and x_n if

$$d(x_{i-1}, x_i) < \varepsilon \quad \text{for } i = 1, 2, \dots, n.$$

The metric space (X, d) is ε -chainable if for each pair (x, y) of its points, there exists an ε -chain joining them.

Interpretation: The space can be connected by chains of small steps of size less than ε .

Examples: Connected metric spaces (e.g., intervals in $\mathbb{R}$, connected subsets of $\mathbb{R}^n$).

(ε, k) – Uniformly Locally Contractive Mappings

Definition

A mapping $T : X \rightarrow X$ is called (ε, k) -uniformly locally contractive if there exists $\varepsilon > 0$ and $k \in [0, 1)$ such that

$$d(x, y) < \varepsilon \implies d(Tx, Ty) \leq k d(x, y) \quad \text{for all } x, y \in X.$$

Key Property: T contracts distances only when they are below the threshold ε .

Comparison:

- **Standard Contraction:** Contracts all distances
- **Locally Contractive:** Contracts only small distances

Edelstein Theorem

Theorem (Edelstein, 1962)

Let (X, d) be a complete ε -chainable metric space and $T : X \rightarrow X$ be an (ε, k) -uniformly locally contractive mapping. Then:

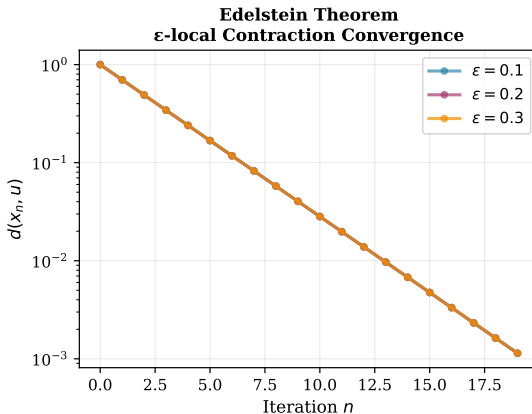
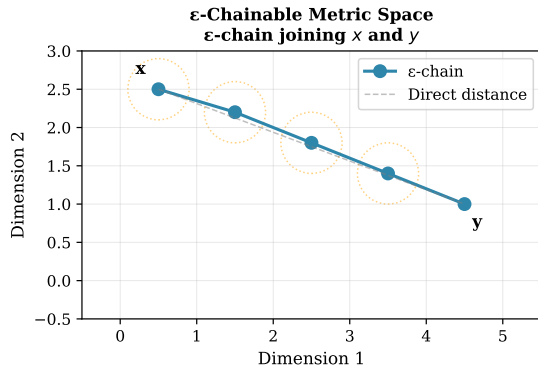
- 1 T has a unique fixed point $u \in X$;
- 2 For any $x_0 \in X$, the sequence $\{T^n x_0\}$ converges to u .

Proof Idea:

- 1 Define a new metric $d_\varepsilon(x, y) = \inf \sum d(x_{i-1}, x_i)$ over all ε -chains
- 2 Show that T is a contraction with respect to d_ε
- 3 Apply Banach fixed point theorem

Significance: Generalizes contraction principle to spaces that are only locally contractive rather than globally contractive.

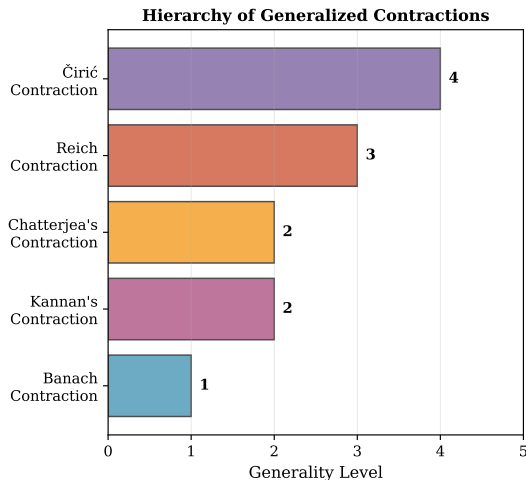
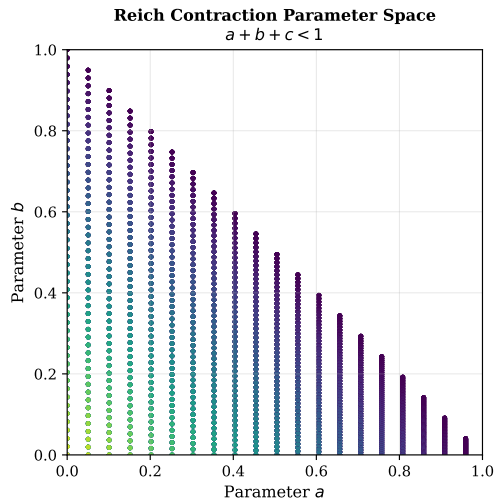
Illustration: ε – Chainable Spaces and Edelstein



Left: An ε –

chain connects distant points through intermediate steps. **Right:** Convergence under locally contractive mappings with different localization parameters.

Hierarchy of Contractive Mappings



Different contraction types form a hierarchy of generality. More general conditions allow weaker assumptions on the mapping T while still guaranteeing a unique fixed point.

Summary: Generalized Contraction Types

Type	Condition	Year
Banach	$d(Tx, Ty) \leq \lambda d(x, y)$	1922
Kannan	$d(Tx, Ty) \leq r[d(Tx, y) + d(Ty, x)], r < 1/2$	1968
Chatterjea	$d(Tx, Ty) \leq r[d(Tx, y) + d(Ty, x)], r < 1/2$	1972
Meir-Keeler	$\varepsilon \leq d(x, y) < \varepsilon + \delta \Rightarrow d(Tx, Ty) < \varepsilon$	1969
Reich	$d(Tx, Ty) \leq ad(x, y) + bd(x, Tx) + cd(y, Ty)$	1971
Čirić	Above + $e[d(x, Ty) + d(y, Tx)], a + b + c + 2e < 1$	1971
Hardy-Rogers	5 parameters: $\sum a_i < 1$	1973
Zamfirescu	Max of three expressions	1972
Berinde	$d(Tx, Ty) \leq rd(x, y) + Ld(y, Tx)$	2004

Numerical Example: Contraction Comparison

```
1 # Numerical comparison of convergence rates
2 import numpy as np
3
4 def iterate_T(x0, k, n_iter=20):
5     """Simulate  $T^n(x_0) = k \cdot x_0$  converging to 0"""
6     return [x0 * k**n for n in range(n_iter)]
7
8 # Compare different contraction constants
9 k_values = [0.3, 0.6, 0.9]
10 x0 = 1.0
11
12 for k in k_values:
13     iterates = iterate_T(x0, k, 15)
14     print(f"k = {k}:")
15     for n, x_n in enumerate(iterates[:5]):
16         print(f"    x_{n} = {x_n:.6f}")
17
18 # Estimate order of convergence
19 rates = [np.log(iterates[n+1]/iterates[n])
20         for n in range(len(iterates)-1) if iterates[n] > 1e-10]
21 print(f"    Convergence rate: {np.mean(rates):.4f}")
22 print()
```


Applications of Generalized Contractions

- ➊ **Differential Equations:** Many existence and uniqueness theorems for ODEs rely on contraction principles
- ➋ **Integral Equations:** Fixed point methods solve Volterra and Fredholm equations
- ➌ **Optimization:** Iterative methods for minimization use contractive principles
- ➍ **Numerical Analysis:** Newton-Raphson and related methods use contraction properties
- ➎ **Economics:** Equilibrium problems in game theory and markets
- ➏ **Dynamical Systems:** Stability analysis and bifurcation theory

Independence of Contraction Types

Theorem (Rhoades, 1977)

*The following contractive conditions are **independent**:*

- *Banach contraction (BC)*
- *Kannan contraction (KC)*
- *Chatterjea contraction (CHC)*

meaning that none implies the others in general.

Implications:

- Each contraction type captures different aspects of contractive behavior
- The choice of contraction depends on the problem structure
- Generality comes at the cost of more complex conditions

Open Question: How can we systematically search for the “right” contraction type for a given problem?

Key Theorems Summary

Theorem	Conclusion
Banach Contraction	Unique fixed point exists
Kannan Fixed Point	Unique fixed point (non-continuous)
Chatterjea Fixed Point	Unique fixed point (non-continuous)
Meir-Keeler	Contractive implies unique fixed point
Reich Fixed Point	Unique fixed point (mixed distances)
Čirić Fixed Point	Unique fixed point (most general linear)
Hardy-Rogers Fixed Point	Unique fixed point (5 parameters)
Bessaga Converse	Metric can be constructed from iterates
Edelstein Theorem	Unique fixed point in chainable spaces

Common Theme: All theorems guarantee existence of a unique fixed point under weaker contractive conditions than Banach's.

Weakly Contractive Mappings (Banach Spaces)

Definition (Weakly Contractive Mapping - Banach Space)

Let X be a Banach space. A self-map $T : X \rightarrow X$ is *weakly contractive* if for every $x, y \in X$,

$$\|Tx - Ty\| \leq \|x - y\| - \psi(\|x - y\|),$$

where $\psi : [0, \infty) \rightarrow [0, \infty)$ is continuous, nondecreasing, and satisfies:

- 1 $\psi(0) = 0$;
- 2 $\psi(t) > 0$ for $t > 0$;
- 3 $\lim_{t \rightarrow \infty} \psi(t) = \infty$.

Examples of ψ :

- $\psi(t) = at$ for $a \in (0, 1)$ (linear)
- $\psi(t) = \frac{t^2}{t+1}$ (nonlinear)
- $\psi(t) = \ln(t+1)$ (logarithmic)

Weakly Contractive Mappings (Metric Spaces)

Definition (Weakly Contractive Mapping - Metric Space)

Let (X, d) be a metric space. A self-map $f : X \rightarrow X$ is *weakly contractive* if for all $x, y \in X$,

$$d(fx, fy) \leq d(x, y) - \psi(d(x, y))$$

where ψ satisfies the conditions above.

Key Difference from Banach:

- Weakly contractive: $d(fx, fy) \leq d(x, y) - \psi(d(x, y))$ (strict improvement)
- Contractive: $d(Tx, Ty) \leq \lambda d(x, y)$ with $\lambda < 1$ (linear ratio)

Advantage: The decay function ψ can be arbitrary (not necessarily linear), providing much flexibility.

Theorems for Weakly Contractive Mappings

Theorem (Alber-Guerre-Delabriere)

Let (X, d) be a complete metric space and $f : X \rightarrow X$ be a weakly contractive map. Then f has a unique fixed point $u \in X$.

Theorem (Harjani-Sadarangani, 2009)

Let $(X, \leq)$ be a partially ordered set with a metric d such that (X, d) is complete. Let $f : X \rightarrow X$ be continuous and nondecreasing such that

$$d(fx, fy) \leq d(x, y) - \psi(d(x, y)) \quad \text{for } x \geq y$$

If there exists $x_0 \in X$ with $x_0 \leq f(x_0)$, then f has a fixed point.

Extension: Weakly contractive mappings can be combined with ordering structures to handle more complex spaces.

Summary and Main Takeaways

- 1 **Classical Contraction:** Banach's principle requires global Lipschitz property
- 2 **Generalized Contractions:** Kannan, Chatterjea, Reich, Ćirić, Hardy-Rogers progressively generalize the contraction condition
- 3 **Local Contractions:** Meir-Keeler and Edelstein theorems handle locally contractive mappings
- 4 **Converse Result:** Bessaga shows the principle is complete — iterates determine the metric
- 5 **Modern Theory:** Weakly contractive mappings provide nonlinear decay guaranteeing convergence
- 6 **Applications:** These results apply to differential equations, integral equations, optimization, and dynamical systems

Further Reading

- Pathak, H. K. (2018). *An Introduction to Nonlinear Analysis and Fixed Point Theory*. Springer.
- Kirk, W. A., & Sims, B. (Eds.). (2001). *Handbook of Metric Fixed Point Theory*. Kluwer Academic Publishers.
- Berinde, V. (2007). *Iterative Approximation of Fixed Points*. Springer.
- Khamsi, M. A., & Kirk, W. A. (2011). *An Introduction to Metric Spaces and Fixed Point Theory*. John Wiley & Sons.

Key Theorems at a Glance I

- **Banach (1922):** $d(Tx, Ty) \leq \lambda d(x, y) \Rightarrow$ unique fixed point
- **Kannan (1968):** $d(Tx, Ty) \leq r[d(Tx, y) + d(Ty, x)]$, $r < 1/2 \Rightarrow$ unique fixed point
- **Chatterjea (1972):** Mixed-distance condition $\Rightarrow$ unique fixed point
- **Meir-Keeler (1969):** Epsilon-delta condition $\Rightarrow$ contractive $\Rightarrow$ unique fixed point
- **Reich (1971):** Linear combination of mixed distances $\Rightarrow$ unique fixed point
- **Čirić (1971):** Extends Reich with additional mixed terms
- **Hardy-Rogers (1973):** Five-parameter generalization
- **Rhoades (1977):** Basic contractions are independent
- **Bessaga (1959):** Metric construction from iterates
- **Edelstein (1962):** Local contractions in chainable spaces
- **Alber-Guerre-Delabriere:** Weakly contractive mappings
- **Harjani-Sadarangani (2009):** Weakly contractive in ordered spaces